USB-PD base board: 20 V from any PD charger that offers 20 V, switched on only after the contract.

CH224A asks for 20 V (R1 120k on CFG1). PG is open-drain, 32 V rated, low once the requested voltage is in place.

Q1 stays off until PG: R2/C2 hold the gate at the source. When PG pulls low, R2/R3 set Vgs = −10 V,

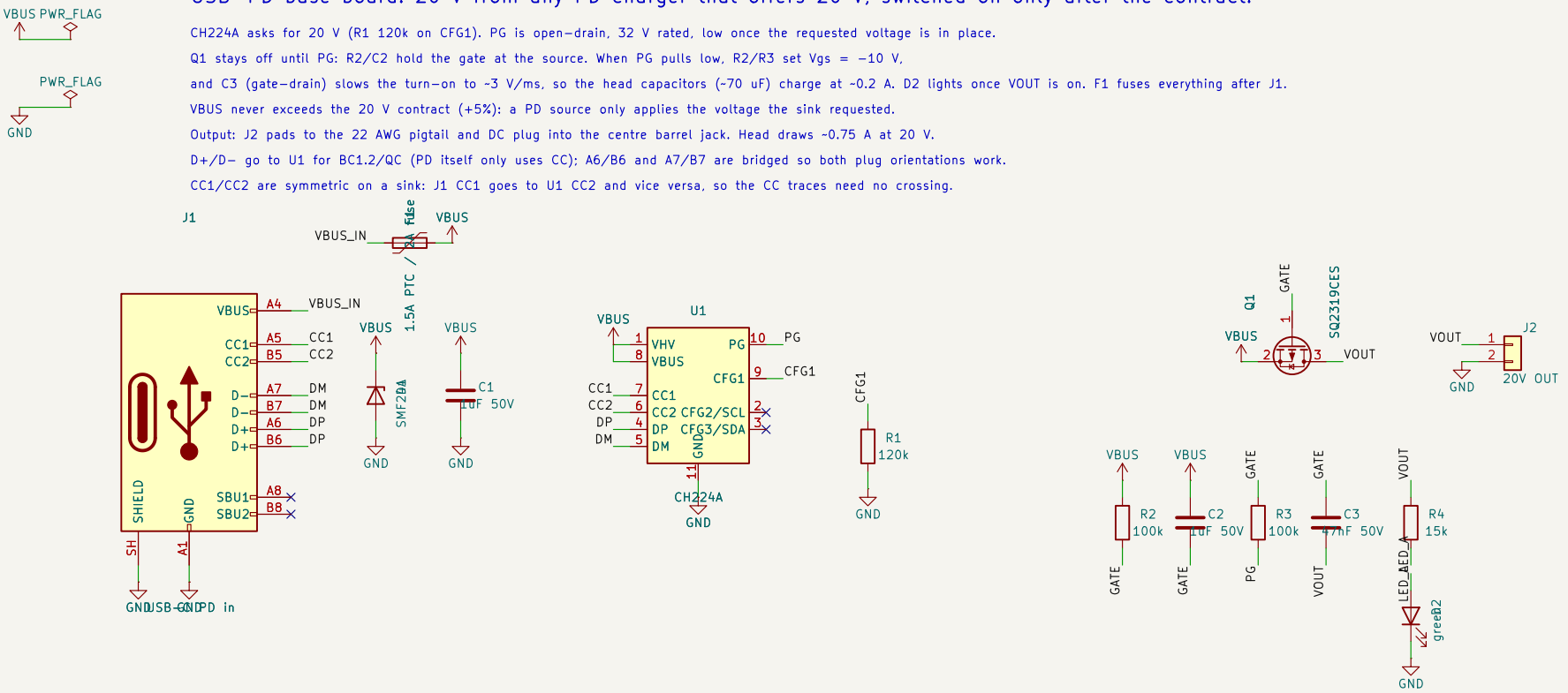
and C3 (gate-drain) slows the turn-on to ~3 V/ms, so the head capacitors (~70 uF) charge at ~0.2 A. D2 lights once VOUT is on. F1 fuses everything after J1.

VBUS never exceeds the 20 V contract (+5%): a PD source only applies the voltage the sink requested.

Output: J2 pads to the 22 AWG pigtail and DC plug into the centre barrel jack. Head draws ~0.75 A at 20 V.

D+/D− go to U1 for BC1.2/QC (PD itself only uses CC); A6/B6 and A7/B7 are bridged so both plug orientations work.

CC1/CC2 are symmetric on a sink: J1 CC1 goes to U1 CC2 and vice versa, so the CC traces need no crossing.



H1 M2 H2 M2 H3 M2 H4 M2

CH224A (20 V request) + AO3407A output switch gated by PG
20 V USB-PD sink for the base pocket, feeds the head over the centre barrel jack

Sheet: /
File: OTL-Base-PD.kicad_sch

Title: Open Task Light USB-PD Base Board

Size: A4	Date: 2026-10-10	**Rev: 1.0**
KiCad E.D.A. 10.0.3		Id: 1/1